

Multi-Hop (Entry/Exit Separation)

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Volume 2 (Data Plane) nano-detail specification. Deepens the **Multi-Hop (nested WireGuard)** section of the pass-1 data-plane overview [01-data-plane §11.1]. SPEC-ONLY: this document describes *what to build* (Rust traits, structs, enums, fns, wire byte layouts, state machines, error taxonomy, config knobs, edge cases, security, performance, tests). It does not ship the product. Source evidence cited inline by id, e.g. [04_ARCH §3.5], [01-DP §11.1], [SYNTHESIS §3 D6], [research-pki_pq_nat §3].

0. Position, scope, and what this document owns

HelixVPN multi-hop generalises the original "chain two VPSes" design into a client-driven **nested-WireGuard tunnel chain** [04_ARCH §3.5, 01-DP §11.1]. The canonical 2-hop topology:

Client → Gateway-Entry → Gateway-Exit → { public internet | connector LAN }

The defining privacy property — the **Mullvad multi-hop property** [04_ARCH §3.5] — is **entry/exit separation**:

Hop	Knows	Does NOT know
Entry (first gateway)	the client's real source IP; that <i>some</i> exit follows	the final destination, the plaintext
Exit (last gateway)	the final destination; the exit-leg ciphertext source	

Hop	Knows	Does NOT know
		the client's real source IP, the plaintext

No single relay simultaneously sees *who* the client is **and** *what* it is talking to. This is achieved structurally, by **nesting two independent WireGuard sessions** so each hop can only strip its own layer, never the inner one [04_ARCH §3.5].

This document owns: the `helix-core/src/hop_chain.rs` module — the nested-WG data structures, the encapsulation/decapsulation pipeline, the per-hop key model, the hop-chain state machine, the multi-hop RouteMap consumption, the MTU/overhead budget for chains, the error taxonomy, config knobs, edge cases, and the §11.4.169 test points. It is the Phase-2 work item `helix-core/src/hop_chain.rs` from the build checklist [01-DP §12].

This document does NOT own (cross-references): the Transport trait and its impls (owned by [01-DP §3] — each hop reuses an ordinary Transport *unchanged*); the `helix-wg` boringtun wrapper [01-DP §4]; the WatchNetworkMap wire contract that *streams* the multi-hop chain (doc 03); the control-plane chain-selection / jurisdiction policy that *decides* which Entry+Exit pair to assign (doc 02); the post-quantum PSK layer (security doc); the kill-switch/DNS-leak interaction (security doc, referenced at §11).

0.1 Non-negotiable multi-hop invariants

These extend the data-plane invariants I1-I6 [01-DP §0.1]; an implementation violating any is a release blocker.

#	Invariant	Source
M1	No new wire protocol. A hop chain is N independent WireGuard sessions over N independent Transports; the bytes on each leg are exactly what that leg's Transport already carries (I1/I2 preserved per-leg).	[01-DP §11.1], [04_ARCH §3.5]
M2	Per-hop keys are independent. The client holds a <i>distinct</i> Curve25519 keypair and a <i>distinct</i> WG session for every hop. No key is shared across hops.	[04_ARCH §3.5], [research-pki_pq_nat §1.1]
M3	Strict layering: each hop strips exactly one layer. Entry decrypts the outer WG session and forwards the <i>still-encrypted</i> inner datagram; only Exit can decrypt the inner session. Entry never possesses the inner key.	[04_ARCH §3.5]

#	Invariant	Source
M4	Entry-blindness / Exit-blindness by construction , not by policy: Entry's AllowedIPs for the client peer permit only the Exit overlay address; Exit's view of the source is the Entry leg, never the client.	[04_ARCH §3.5], [01-DP §7]
M5	Fail-static per hop. If the control plane is down, an established chain keeps forwarding (I3); a single-hop failure tears down the <i>whole chain</i> (no silent half-tunnel) and surfaces Down.	[01-DP §0.1 I3]
M6	Jurisdiction separation is an input, not a guarantee the data plane can verify. The data plane faithfully chains whatever Entry/Exit the control plane assigns; it cannot itself attest the two gateways are in different jurisdictions (that is doc-02 policy). The data plane MUST expose the assigned jurisdiction labels in status so the UX/policy layer can assert the property.	[SYNTHESIS §7], §11.4.6

1. Conceptual model — nested WireGuard

A single-hop tunnel is one WG session inside one Transport:

```
[ inner IP packet ] —WG_exit?— no, single hop: [ IP ] —WG(gw)→
[ Transport ] → wire
```

A 2-hop chain composes **two** WG sessions where the *inner* session's encrypted datagram is the *plaintext payload* of the *outer* session [04_ARCH §3.5, 01-DP §11.1]:

```

      ┌───────────────────────────────────────────────────────────────────────────────────┐ what the CLIENT does on
egress ────────────────────────────────────────────────────────────────────────────────────┘
      |
      | user IP pkt
      |
      | ─① WG-Exit encrypt (client↔Exit session, key pair K_exit) →
      | Ed = WG_exit(IP)
      |
      |      Ed is an opaque WireGuard transport-data message
      |      addressed to the Exit overlay IP
      |
      | ─② treat Ed as the *inner IP payload* destined for the Exit
      | overlay address, then
      |   WG-Entry encrypt (client↔Entry session, key pair K_entry)
      | → Ee = WG_entry(Ed)

```

|
↳ Ee handed to the ENTRY-leg Transport.send() (e.g. masquerade) → wire to Entry:443

On the return path the client reverses: the Entry Transport yields Ee, WG-Entry decrypts to Ed, WG-Exit decrypts to the user IP packet.

The gateways do the middle unwrapping:

ENTRY gateway: Transport.recv() → Ee → WG_entry.decrypt → Ed
(still WG ciphertext)
→ route Ed toward Exit over the Entry↔Exit
Transport (Entry CANNOT read inside Ed; M3)
EXIT gateway: Transport.recv() → Ed → WG_exit.decrypt → user IP
packet
→ forward to {internet | connector} per Exit
policy

Key insight (M1): from each Transport's point of view nothing changed. The Entry leg carries opaque WG datagrams (Ee); the Entry↔Exit leg carries opaque WG datagrams (Ed). The nesting lives entirely in helix-core's encrypt/decrypt ordering — **not** in any new framing on the wire [01-DP §11.1]. This is why §11.1 says "the transport layer is unchanged."

1.1 Generalisation to N hops

The 2-hop case is the MVP target [04_ARCH §3.5]. The data structures generalise to an ordered chain [H_0 =Entry, H_1 , ..., H_{n-1} =Exit]; egress encrypts **innermost-first** (Exit session applied first, Entry session applied last) and the outermost layer is handed to H_0 's transport. Each intermediate hop strips one layer and forwards. The spec implements and tests **$n \leq \text{MAX_HOPS}$ (default 4)**; $n=2$ is the only configuration with a Phase-2 acceptance gate, $n=3/4$ are validated by unit/integration tests only and gated behind config (HopChainConfig.max_hops). MAX_HOPS = 4 is a denial-of-amplification ceiling, not a protocol limit (UNVERIFIED that any source mandates a specific ceiling — chosen here as a safe default per §11.4.6; tune in Phase 2 from latency budget §10).

2. Module layout & data model (helix-core/src/hop_chain.rs)

The pass-1 sketch [01-DP §11.1] is:

```

pub struct HopChain { pub hops: Vec<Hop> }           // [Entry,
                Exit] for 2-hop
pub struct Hop { pub wg: WgPeer, pub transport: Box<dyn
                Transport> }

```

This section nano-details it. The full type set:

```

// helix-core/src/hop_chain.rs
use crate::status::TunnelStatus;
use crate::wg::{WgPeer, WgVerdict, WgError};           //
                [01-DP §4]
use helix_transport::{Transport, TransportConfig, TransportError,
                dial};
use helix_route::RouteMap;                             //
                [01-DP §6.2]
use bytes::{Bytes, BytesMut};
use std::net::IpAddr;
use std::time::{Duration, Instant};

/// An ordered nested-WireGuard tunnel chain. `hops[0]` is ENTRY
    (outermost on the wire),
/// `hops[len-1]` is EXIT (innermost crypto layer, owns the user
    IP packet). Invariant M2:
/// every Hop carries a DISTINCT WgPeer (distinct key material).
pub struct HopChain {
    pub hops:
        Vec<Hop>,           // 1 = single hop (degenerate), 2 =
        Entry/Exit (MVP), ≤ MAX_HOPS
    state: HopChainState,    // §5 state machine
    cfg: HopChainConfig,     // §7 knobs
    scratch: BytesMut,       // reused egress/ingress
        encryption buffer (no per-packet alloc)
    metrics:
        HopChainMetrics,    // aggregate counters only (I5, no
        per-packet durable state)
}

/// One leg of the chain. The `wg` session and the `transport` are
    BOTH specific to this leg.
pub struct Hop {
    pub idx:
        u8,                 // position in the chain, 0 =
        Entry
    pub role: HopRole,       // Entry | Intermediate |
        Exit
    pub wg: WgPeer,          // distinct client↔hop
        WireGuard session (M2)
    pub transport: Box<dyn Transport>, // this leg's L2 carrier
        (may differ per leg – D6 §8)
    pub overlay: IpAddr,     // this hop gateway's
        overlay address (next-inner dst)
    pub jurisdiction: Jurisdiction, // label from the map,
        surfaced for M6 assertion
    health: HopHealth,
}

```

```

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub enum HopRole { Entry, Intermediate, Exit }

/// Opaque label assigned by the control plane (doc 02). The data
plane never INTERPRETS it
/// beyond equality/inequality for the M6 distinct-jurisdiction
surface; it is a string id
/// (ISO-3166 alpha-2 or operator-defined region tag), NOT a trust
assertion (M6, §11.4.6).
#[derive(Clone, Debug, PartialEq, Eq)]
pub struct Jurisdiction(pub String);

#[derive(Clone, Debug, Default)]
struct HopHealth {
    rtt_ewma_ms: Option<u32>,
    last_rcv_age_ms: u64,
    handshake_ok: bool,
}

#[derive(Clone, Debug, Default)]
struct HopChainMetrics {
    chains_established: u64,
    chain_tear_downs: u64,
    egress_datagrams: u64,
    ingress_datagrams: u64,
    oversize_drops: u64, // §6 MTU
    per_hop_handshake_failures: [u32; 8], // indexed by hop idx;
    aggregate, no per-flow state (I5)
}

```

HopChain is owned by the Orchestrator [01-DP §5.2] in place of the single (WgPeer, Box<dyn Transport>) pair when multi-hop is active. A single-hop tunnel is the degenerate HopChain { hops: vec! [Hop{role: Exit, idx: 0, ...}] }, so the orchestrator's two data loops have ONE code path (chain of length 1 or N) — this is the key simplification that keeps multi-hop from being a second forwarding engine.

3. The encapsulation pipeline (egress + ingress)

3.1 Egress — innermost-first encryption

```

impl HopChain {
    /// Outbound: a plaintext IP packet from the TUN → the
    outermost (Entry-leg) datagram,
    /// then hand it to hops[0].transport.send(). Innermost (Exit)
    WG layer applied FIRST.

```

```

/// Returns Ok(()) on send, or a HopChainError that the
orchestrator maps to status.
pub async fn egress(&mut self, ip_pkt: &[u8]) -> Result<(),
HopChainError> {
    // ① start with the user IP packet as the innermost
plaintext
    let mut layer: Bytes =
        self.encrypt_exit_layer(ip_pkt)?;           // WG_exit(IP)
        = Ed
    // ② wrap with each preceding hop, Exit-1 down to Entry
(indices len-2 .. 0)
    for i in (0..self.hops.len() - 1).rev() {
        layer = self.encrypt_hop_layer(i,
        &layer)?;           // WG_i(layer)
    }
    // ③ outermost layer = Ee → Entry-leg transport
    self.metrics.egress_datagrams += 1;
    self.hops[0].transport
        .send(layer)
        .await
        .map_err(|e| HopChainError::Leg { hop: 0, src:
        e.into() })
}

/// WG-encrypt `inner` for hop `i`'s session. `inner` is
treated as the IP payload that
/// WG_i protects (for i = Exit this is the real user packet;
for i < Exit it is the
/// next-inner hop's ciphertext Ed – opaque to hop i's
session, M3).
fn encrypt_hop_layer(&mut self, i: usize, inner: &[u8]) ->
Result<Bytes, HopChainError> {
    self.scratch.clear();
    self.scratch.resize(inner.len() + WG_DATAGRAM_OVERHEAD,
    0);
    match self.hops[i].wg.handle_tun_out(inner, &mut
    self.scratch) { // [01-DP §4]
        WgVerdict::WriteToTransport(buf) =>
        Ok(Bytes::copy_from_slice(buf)),
        WgVerdict::Nothing => Err(HopChainError::HopNotReady
        { hop: i as u8 }), // handshake pending
        WgVerdict::Err(e) => Err(HopChainError::Wg { hop: i
        as u8, src: e }),
        WgVerdict::WriteToTun(_) => unreachable!("tun_out
        never yields tun_in"),
    }
}

#[inline]
fn encrypt_exit_layer(&mut self, ip_pkt: &[u8]) ->
Result<Bytes, HopChainError> {
    self.encrypt_hop_layer(self.hops.len() - 1, ip_pkt)
}
}

```

WG_DATAGRAM_OVERHEAD is the per-layer WireGuard transport-data overhead: **32 bytes** (4-byte type/reserved + 4-byte receiver index + 8-byte counter/nonce + 16-byte Poly1305 tag) [01-DP §10 derivation]. Each nesting level adds this, so a 2-hop chain pays **2 × 32 = 64 bytes** of WG overhead before any transport overhead (the MTU consequence is §6).

3.2 Ingress — outermost-first decryption (client side)

```
impl HopChain {
    /// Inbound on the client: a datagram arrived on the Entry-leg transport (Ee).
    /// Peel Entry → ... → Exit; the Exit layer yields the user IP packet for the TUN.
    /// Handshake replies on any layer are re-emitted out that hop's transport (M5 keepalive).
    pub async fn ingress(&mut self) -> Result<IngressOutcome, HopChainError> {
        let outer = self.hops[0].transport
            .recv().await
            .map_err(|e| HopChainError::Leg { hop: 0, src: e.into() })?;
        self.metrics.ingress_datagrams += 1;
        self.peel(0, outer)
    }

    /// Recursively strip layer `i`. WriteToTransport at layer i = a WG handshake/keepalive
    /// reply for hop i → send it back out hop i's transport (does NOT continue peeling).
    fn peel(&mut self, i: usize, datagram: Bytes) -> Result<IngressOutcome, HopChainError> {
        self.scratch.clear();
        self.scratch.resize(datagram.len(), 0);
        match self.hops[i].wg.handle_transport_in(&datagram, &mut self.scratch) {
            WgVerdict::WriteToTun(plain) if i == self.hops.len() - 1 => {
                /// innermost: this is the real user IP packet
                Ok(IngressOutcome::ToTun(Bytes::copy_from_slice(plain)))
            }
            WgVerdict::WriteToTun(inner) => {
                /// an intermediate layer decrypted to the NEXT-inner ciphertext → keep peeling
                self.peel(i + 1, Bytes::copy_from_slice(inner))
            }
            WgVerdict::WriteToTransport(reply) => {
                /// handshake/keepalive reply for hop i → re-send out hop i's transport
                Ok(IngressOutcome::HopReply { hop: i as u8, datagram: Bytes::copy_from_slice(reply) })
            }
        }
    }
}
```



```

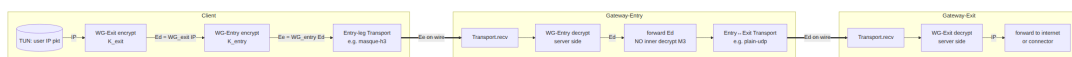
    }
    WgVerdict::Nothing =>
    Ok(IngressOutcome::Consumed), // e.g. cookie/decoy
    WgVerdict::Err(e) => Err(HopChainError::Wg { hop: i
as u8, src: e }},
    }
}
}

pub enum IngressOutcome {
    ToTun(Bytes), // deliver to
                    the OS TUN
    HopReply { hop: u8, datagram:
        Bytes }, // re-send out hop's transport
                (orchestrator)
    Consumed, // handled
              internally, nothing to forward
}

```

Edge case — peel/encrypt asymmetry on intermediate hops. On the *client*, the user drives `handle_tun_out` (encrypt) for the egress chain and `handle_transport_in` (decrypt) for ingress. The *gateways* (Entry/Intermediate) run a DIFFERENT loop: they decrypt the arriving outer layer with their *server-side* `WgPeer` for the client, then forward the resulting inner ciphertext over the next leg WITHOUT decrypting it (M3) — they do not own the inner key. The gateway forwarding path is specified in §4; peel above is the client view.

3.3 Packet-flow diagram (2-hop egress)



4. Gateway (Entry / Intermediate) forwarding contract

Note. The gateway *role* in a hop chain runs on the helix-edge binary [01-DP §11.2, D5] using the **same** helix-transport + helix-wg crates. This section specifies the *forwarding contract* the edge must satisfy to participate in a chain; the edge binary's full spec is doc 01-DP §11.2 / the edge volume. It is included here because the contract is the data-plane half of entry/exit separation.

An Entry (or any non-Exit) gateway, for a given client chain, holds a **server-side** `WgPeer` for the client's per-hop public key. On a datagram arriving from the client-facing transport:

```

// helix-edge: chain-forwarding (one hop's worth)
pub struct ChainForwarder {
    inbound_wg: WgPeer,           // server side of
    // client↔this-hop WG session
    inbound_tx: Box<dyn Transport>, // carrier from the
    // previous node (client or prev hop)
    next_tx: Box<dyn Transport>,   // carrier to the NEXT
    // node (next hop or Exit)
    // NO key for any inner layer – M3 enforced structurally (the
    // type holds none)
}

impl ChainForwarder {
    pub async fn pump(&mut self, scratch: &mut [u8]) ->
    Result<(), ForwardError> {
        let outer = self.inbound_tx.recv().await?; // Ee
        match self.inbound_wg.handle_transport_in(&outer,
            scratch) {
            WgVerdict::WriteToTun(inner_ciphertext) => {
                // `inner_ciphertext` is Ed – STILL WireGuard-
                // encrypted for the Exit session.
                // M3: this node CANNOT decrypt it; it only
                // forwards toward the next hop.

                self.next_tx.send(Bytes::copy_from_slice(inner_ciphertext)).await?;
            }
            WgVerdict::WriteToTransport(reply) => {

                self.inbound_tx.send(Bytes::copy_from_slice(reply)).await?; //
                // handshake reply
            }
            WgVerdict::Nothing => {} //
            // cookie / decoy / keepalive consumed
            WgVerdict::Err(e) => return Err(ForwardError::Wg(e)),
        }
        Ok(())
    }
}

```

Why the inner layer is opaque to Entry (M3, the security crux): the inbound_wg session decrypts the *outer* WG transport-data message. WireGuard's AllowedIPs for the client peer on the Entry gateway are set to **exactly the Exit overlay address** [M4, 01-DP §7]. The decrypted "IP packet" Entry recovers is Ed — but Ed is itself a WireGuard message, indistinguishable from random bytes to anyone without K_{exit} . Entry forwards it to the Exit endpoint over the Entry↔Exit transport because the routing map says the Exit overlay is reachable via that leg. Entry has no K_{exit} (it was never given one — M2), so it provably cannot learn the destination [04_ARCH §3.5]. Symmetrically, the Exit gateway's inbound_wg is the client↔Exit server session; it sees Ed arriving from the *Entry* leg, so the source it observes is Entry, not the client.

```
pub enum ForwardError {
    Transport(#[from] TransportError),
    Wg(WgError),
    NextLegDown,
}
```

5. Hop-chain state machine

The chain has a single lifecycle that gates egress/ingress and drives TunnelStatus [01-DP §5.2]. Egress is permitted only in Established; any hop failing transitions the WHOLE chain (M5 — no half-tunnel).

```
#[derive(Clone, Debug, PartialEq, Eq)]
pub enum HopChainState {
    Idle,
    Dialing { hop: u8 }, // dial()ing hop `hop`'s
                        transport, Entry→Exit order
    Handshaking { hop: u8 }, // WG handshake for hop
                        `hop` in flight
    Established { since: Instant }, // all hops handshook;
                        forwarding live
    Rekeying { hop: u8 }, // a hop is mid-rekey
                        (still forwarding on old keys)
    TearingDown { reason: TeardownReason },
    Failed { hop: u8, reason: TeardownReason },
}

#[derive(Clone, Debug, PartialEq, Eq)]
pub enum TeardownReason {
    ClientRequested,

    HopHandshakeTimeout, // a hop never completed WG
                        handshake within budget
    HopTransportDown, // a leg's Transport returned Closed/
                        EndpointBlocked
    MapWithdrew, // control plane removed this chain
                from the RouteMap
    Oversize, // §6 unrecoverable MTU violation
    JurisdictionViolation, // M6: assigned Entry==Exit
                        jurisdiction with strict policy on (§7)
}
```

5.1 Establishment order (Entry-first, then inner)

The chain is brought up **Entry-first, innermost-last**: the Entry transport + Entry WG handshake must succeed before the Exit handshake can traverse it (the Exit handshake packets are themselves nested inside the Entry session) [derived from §1 nesting; 04_ARCH §3.5].

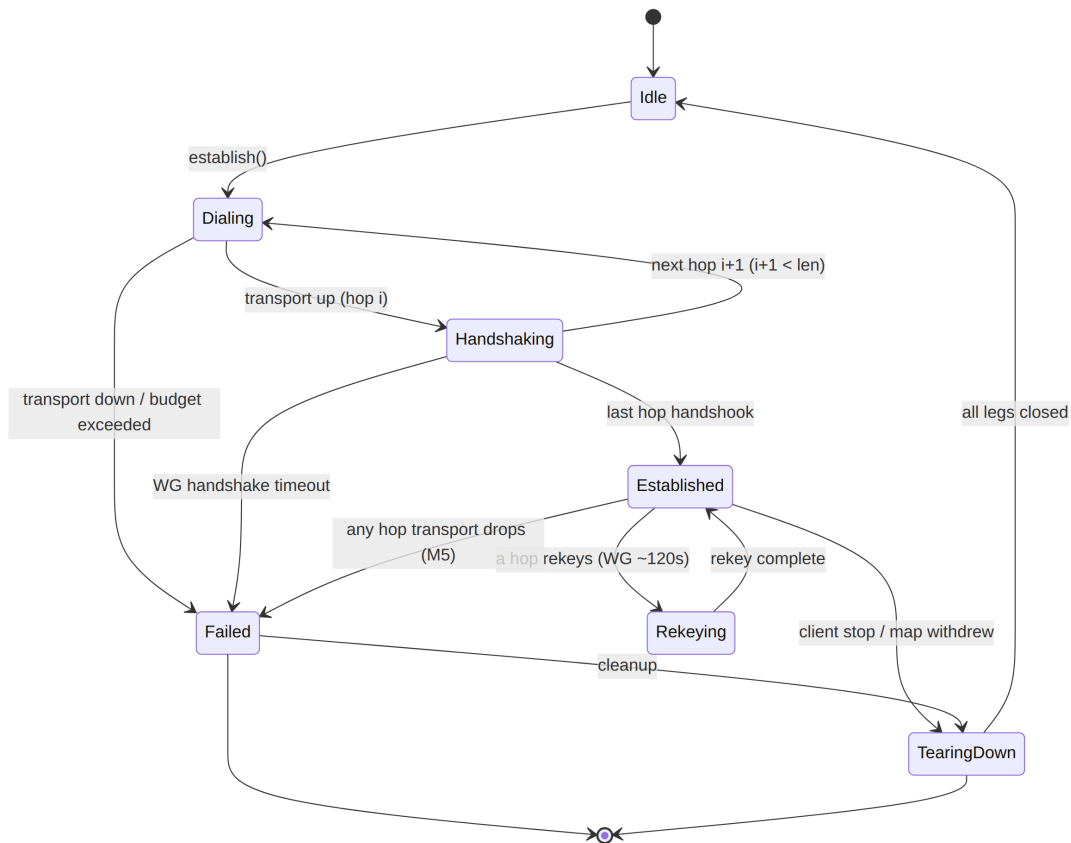
```

impl HopChain {
    /// Build the whole chain from a multi-hop RouteMap (§8).
    /// Dials + handshakes Entry→Exit.
    /// Emits TunnelStatus deltas via the orchestrator's broadcast
    /// (caller wires the channel).
    pub async fn establish(
        &mut self,
        on_status: &mut dyn FnMut(TunnelStatus),
    ) -> Result<(), HopChainError> {
        on_status(TunnelStatus::Connecting);
        for i in 0..self.hops.len() {
            self.state = HopChainState::Dialing { hop: i as u8 };
            // hop 0 dials the real network; hops > 0 "dial" by
            handshaking THROUGH the
            // already-established outer hops (their transport is
            a virtual leg over the chain).
            self.dial_hop(i).await
                .map_err(|e| { self.state = HopChainState::Failed
                    { hop: i as u8,
                      reason: TeardownReason::HopTransportDown };
                    e })?;
            self.state = HopChainState::Handshaking { hop: i as
                u8 };
            on_status(TunnelStatus::Handshaking);
            self.handshake_hop(i).await
                .map_err(|e| { self.state = HopChainState::Failed
                    { hop: i as u8,
                      reason:
                        TeardownReason::HopHandshakeTimeout }; e })?;
        }
        self.state = HopChainState::Established { since:
            Instant::now() };
        self.metrics.chains_established += 1;
        on_status(TunnelStatus::Connected {
            transport:
                self.entry_kind_label(), // e.g. "masque-
h3→plain-udp"
            rtt_ms: self.aggregate_rtt_ms(), // §10
sum of per-hop RTTs
        });
        Ok(())
    }
}

```

entry_kind_label() returns a chain label of the form "<entry-kind>→...→<exit-kind>" so the status UX shows the full path (e.g. "masque-h3→plain-udp"). aggregate_rtt_ms() is the sum of per-hop RTT EWMA's (latency is additive across a serial chain — §10).

5.2 State diagram



M5 elaboration. From Established, *any* single hop losing its transport or missing WG keepalive past the budget transitions the chain to Failed{hop} and then TearingDown. The orchestrator emits Down{reason} and (per the ladder policy [01-DP §5.3]) MAY re-establish. The chain is NEVER left in a state where some hops forward and others are dead — that would leak (e.g. Entry up but Exit dead → traffic black-holes at Entry, which is also where the client's identity is known). Atomicity of teardown is a CONC/RACE test point (§12).

6. MTU & overhead budget for chains

Each nesting level adds one WG transport-data header+tag (32 B, §3.1) AND the per-leg transport overhead. The **inner WG MTU** the client may use is bounded by the *outermost* leg's `effective_mtu()` minus the cumulative inner-WG overhead [01-DP §10].

For an N-hop chain with hop transports $T_0 \dots T_{n-1}$ (T_0 = Entry leg, the wire-facing one):

```

inner_wg_mtu = T0.effective_mtu() -  $\sum_{i=1..n-1}$ 
WG_DATAGRAM_OVERHEAD
            = T0.effective_mtu() - (n-1) · 32

```

Only the Entry leg's transport overhead applies on the wire (the inner legs are encapsulated *inside* the Entry session, so their transport overhead is paid against the inner budget, not the path MTU). Worked examples (using [01-DP §10] table values):

Chain	To (Entry) effective_mtu()	inner- WG overhead (n-1)·32	inner_wg_mtu	Note
1-hop (single)	plain-udp 1420	0	1420	baseline [01-DP §10]
2-hop udp/udp	plain-udp 1420	32	1388	one extra WG layer
2-hop masque/ udp	masque-h3 1280	32	1248	obfuscated entry leg
3-hop masque/ udp/udp	masque-h3 1280	64	1216	approaches IPv6 floor 1280 caution

```

impl HopChain {
  /// The inner WG MTU the client must clamp the TUN to.
  ///   Recomputed on chain (re)establish
  /// and on any per-leg transport change (ladder escalation on
  /// the Entry leg).
  pub fn inner_wg_mtu(&self) -> u16 {
    let entry = self.hops[0].transport.effective_mtu();
    let inner_overhead = (self.hops.len() as u16 - 1) *
      WG_DATAGRAM_OVERHEAD as u16;

    entry.saturating_sub(inner_overhead).max(MIN_INNER_MTU) //
    MIN_INNER_MTU = 1280 floor
  }
}

pub const WG_DATAGRAM_OVERHEAD: usize = 32; // [01-DP §10]
pub const MIN_INNER_MTU: u16 = 1280; // IPv6 minimum-MTU
    floor [01-DP §10]

```

Rules (extend [01-DP §10] rules 1-3):

1. The TUN MTU = min(inner_wg_mtu(), path-MTU-discovered); a chain that would force inner_wg_mtu() below MIN_INNER_MTU is **rejected at establish** with TeardownReason::Oversize rather than silently clamping below the IPv6 floor (a chain that cannot carry a 1280-byte inner packet is unusable for IPv6).

2. `TransportError::Oversize` bubbling from ANY inner leg maps to `HopChainError::Leg` → `metrics.oversize_drops += 1`; the orchestrator lowers the TUN MTU rather than truncating (§11.4.6 — never silently drop bytes).
 3. The 2-hop MASQUE/UDP MTU penalty is **measured, not assumed**, in the Phase-2 BENCH gate (record MTU/throughput/CPU vs single-hop) [01-DP §10, §12].
-

7. Configuration knobs

```
#[derive(Clone, Debug)]
pub struct HopChainConfig {
    /// Hard ceiling on chain length (denial-of-amplification).
    /// Default 4 (§1.1).
    pub max_hops: u8,
    /// Per-hop WG handshake budget before the hop is declared
    /// failed.
    /// Default 5 s (WireGuard's own REKEY_TIMEOUT lineage;
    /// UNVERIFIED exact value – tune).
    pub hop_handshake_timeout: Duration,
    /// Keepalive-miss budget on an Established hop before M5
    /// teardown. Default 25 s
    /// (~ WG persistent-keepalive 25 s window; [research-
    /// pki_pq_nat baseline]).
    pub hop_keepalive_grace: Duration,
    /// If true, refuse a chain whose Entry.jurisdiction ==
    /// Exit.jurisdiction (M6 strict mode).
    /// Default true for the privacy "multi-hop" feature; an
    /// operator MAY relax for
    /// performance-only chaining (e.g. two hops in one region for
    /// routing, not anonymity).
    pub require_distinct_jurisdiction: bool,
    /// Whether the orchestrator re-establishes the chain on M5
    /// teardown (vs surfacing Down).
    /// Default true (auto-reconnect per the ladder [01-DP §5.3]).
    pub auto_reestablish: bool,
}

impl Default for HopChainConfig {
    fn default() -> Self {
        Self {
            max_hops: 4,
            hop_handshake_timeout: Duration::from_secs(5),
            hop_keepalive_grace: Duration::from_secs(25),
            require_distinct_jurisdiction: true,
            auto_reestablish: true,
        }
    }
}
```

`require_distinct_jurisdiction` is the only place the data plane *asserts* M6: it compares the Jurisdiction labels the map supplied. It cannot verify the labels are truthful (M6, §11.4.6) — it surfaces a `JurisdictionViolation` teardown if `Entry==Exit` under strict mode so the control plane's mis-assignment is loud, not silent.

8. Orchestration via the multi-hop RouteMap (no new wire protocol)

The control plane decides the chain and pushes it via `WatchNetworkMap` (doc 03) as an ordered list of `PeerRoutes`. The data plane consumes it — **no new protocol** (M1). The existing `RouteMap/PeerRoute` types [01-DP §6.2] carry a chain by adding one field:

```
// helix-route/src/map.rs (extends [01-DP §6.2]; the ONLY multi-
// hop addition)
pub struct RouteMap {
    pub self_overlay: IpAddr,
    pub peers: Vec<PeerRoute>,           // policy-filtered
        (need-to-know, I6)
    pub dns: Vec<IpAddr>,
    pub hop_chains: Vec<HopChainSpec>,  // NEW: ordered
        chains the client may select
}

/// A control-plane-assigned chain. The client materialises a
/// HopChain from one of these.
pub struct HopChainSpec {
    pub chain_id: u64,                  // stable id for
        status/telemetry (aggregate, I5)
    pub label: String,                 // human label e.g.
        "Entry: SE Exit: CH"
    pub hops: Vec<HopSpec>,            // ordered Entry →
        Exit
}

pub struct HopSpec {
    pub wg_pubkey: [u8; 32],           // this hop
        gateway's WG public key (M2: distinct)
    pub overlay: IpAddr,               // this hop's
        overlay address
    pub endpoint_candidates: Vec<SocketAddr>, // for the leg's
        Transport.dial() / NAT (§ [01-DP §8])
    pub transport: TransportConfig,    // per-leg
        transport (D6 asymmetric-per-leg [SYNTHESIS §3 D6])
    pub jurisdiction: Jurisdiction,    // label for M6
    pub allowed_ips: Vec<IpNet>,      // what this hop's
        session may carry (M4)
}
```



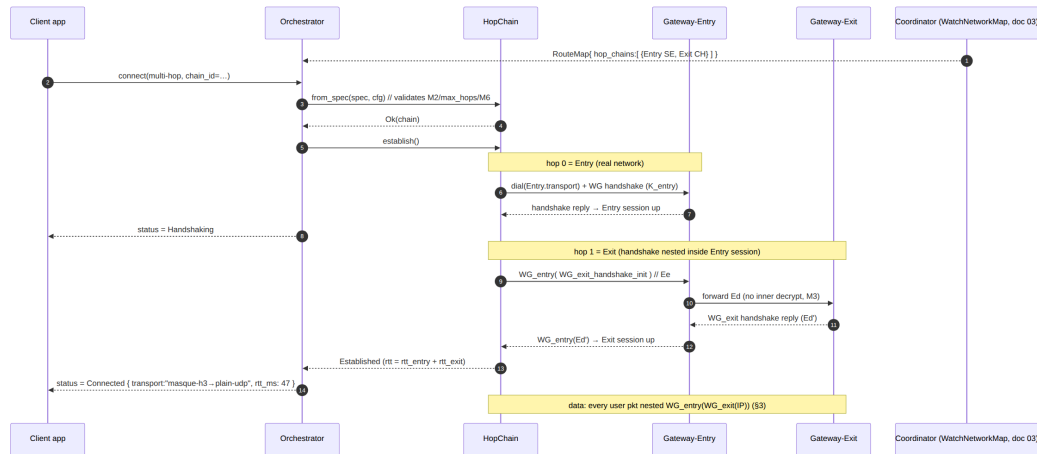
```

impl HopChain {
    /// Build (not yet establish) a chain from a map spec.
    Validates M2 (distinct keys),
    /// max_hops, and (strict mode) distinct jurisdictions BEFORE
    any dial.
    pub fn from_spec(spec: &HopChainSpec, cfg: HopChainConfig) ->
        Result<Self, HopChainError> {
        if spec.hops.is_empty() || spec.hops.len() as u8 >
            cfg.max_hops {
            return Err(HopChainError::BadChainLength { got:
                spec.hops.len() as u8, max: cfg.max_hops });
        }
        /// M2: every hop pubkey distinct.
        let mut seen = std::collections::HashSet::new();
        for h in &spec.hops {
            if !seen.insert(h.wg_pubkey) {
                return Err(HopChainError::DuplicateHopKey);
            }
        }
        /// M6 strict: Entry.jurisdiction != Exit.jurisdiction.
        if cfg.require_distinct_jurisdiction && spec.hops.len()
            >= 2 {
            let entry = &spec.hops.first().unwrap().jurisdiction;
            let exit = &spec.hops.last().unwrap().jurisdiction;
            if entry == exit {
                return Err(HopChainError::JurisdictionNotDistinct
                    {
                        entry: entry.clone(), exit: exit.clone() });
            }
        }
        /// ... construct Hop{} per spec, role = Entry/Intermediate/
        Exit by position ...
        Ok(/* HopChain */ unimplemented_placeholder())
    }
}

```

Reconciliation (push, don't poll) is inherited from [01-DP §6.3]: when the map's hop_chains delta changes (a hop withdrawn, a transport re-pinned, an exit rotated for jurisdiction policy), the reconciler diffs desired-vs-actual and converges WITHOUT restarting the orchestrator — converging the chain in **< 1 s** [01-DP §6.3]. A withdrawn chain triggers TeardownReason::MapWithdrawn. Per-leg transport selection (D6 asymmetric-per-leg) is "free": each HopSpec.transport is an independent TransportConfig, so the Entry leg can be masque-h3 while the Entry↔Exit leg is plain-udp with NO mechanism change [01-DP §11.3, SYNTHESIS §3 D6].

8.1 Establishment sequence diagram (control-plane orchestrated, 2-hop)



9. Error taxonomy

```

#[derive(thiserror::Error, Debug)]
pub enum HopChainError {
    #[error("chain length {got} invalid (1..={max})")]
    BadChainLength { got: u8, max: u8 },
    #[error("duplicate hop key violates per-hop-key invariant M2")]
    DuplicateHopKey,
    #[error("entry and exit share jurisdiction {entry:?}={exit:?} (M6 strict)")]
    JurisdictionNotDistinct { entry: Jurisdiction, exit: Jurisdiction },
    #[error("hop {hop} not ready (handshake pending)")]
    HopNotReady { hop: u8 },
    #[error("leg {hop} transport error: {src}")]
    Leg { hop: u8, #[source] src: TransportError },
    #[error("hop {hop} wireguard error: {src}")]
    Wg { hop: u8, #[source] src: WgError },
    #[error("hop {hop} handshake timed out")]
    HopHandshakeTimeout { hop: u8 },
    #[error("chain torn down: {0:?}")]
    TornDown(TeardownReason),
}

```

Mapping to TunnelStatus [01-DP §5.2] (the orchestrator owns this map):

HopChainError	TunnelStatus emitted	Recovery
	Reconnecting then ladder-	per-leg ladder [01-DP §5.3]

HopChainError	TunnelStatus emitted	Recovery
Leg{HandshakeFailed} / HopHandshakeTimeout during establish	escalate that leg	
Leg{Closed/EndpointBlocked} while Established	Down{reason} (M5 whole-chain)	auto_reestablish
JurisdictionNotDistinct / BadChainLength / DuplicateHopKey	Down{reason} (no auto-retry — config defect)	surface to control plane / operator
HopNotReady (egress before established)	drop egress packet, keep state	retried next packet (no status change)

Honest-failure rule (§11.4.1): every variant fails for a genuine product condition; a chain that *appears* up but black-holes (e.g. Exit silently dead) MUST surface Down within hop_keepalive_grace, never report Connected while not forwarding (that is the §11.4 PASS-bluff at the multi-hop layer).

10. Performance budget

Multi-hop trades latency/throughput for unlinkability. Budgets (Phase-2 BENCH gate [01-DP §12]):

Metric	Single hop	2-hop target	Source / rationale
RTT	baseline r_1	$r_1 + r_2$ (serial sum)	latency is additive along a serial chain; aggregate_rtt_ms() reports the sum
Throughput	$\geq 80\%$ bare link (G1) [01-DP §12]	$\geq$ single-hop minus the second encrypt/decrypt + extra hop transit	UNVERIFIED exact %, set the gate from the Phase-2 measurement, not a literature guess (§11.4.6)
Client CPU/ packet	$1 \times$ WG seal+open	$2 \times$ WG seal+open (nested)	each layer is one ChaCha20-Poly1305 op [01-DP §4]
Inner MTU	1420 (udp)	1388 (udp/udp) ... 1248	§6

Metric	Single hop	2-hop target	Source / rationale
Gateway CPU (Entry)	1 × WG open + forward	(masque/udp) unchanged: 1 × open + forward (M3 — Entry never touches inner crypto)	§4
Convergence on map change	< 1 s [01-DP §6.3]	< 1 s	reconciler is delta-driven, not a restart

Implementation budget rules: (1) the egress/ingress hot path uses the reused scratch BytesMut — **no per-packet heap allocation** per layer (the Bytes::copy_from_slice calls above are the spec's clarity form; the implementation MUST use in-place layering into scratch to hit the CPU budget — a BENCH/MEM test point §12); (2) per-hop RTT EWMA comes from each Transport::health() [01-DP §3.1] — the chain does not run its own probes; (3) no per-flow durable state anywhere (I5) — only the aggregate HopChainMetrics counters.

11. Security considerations

1. **Entry/Exit unlinkability is structural, not policy** (M3/M4). It survives a *malicious* Entry: a compromised Entry gateway still cannot read Ed (no K_exit, M2) and cannot learn the destination. A compromised Exit cannot learn the client IP (it sees Entry). Only an adversary controlling BOTH gateways (or a global passive observer correlating timing) breaks it — the classic multi-hop threat model [04_ARCH §3.5]. The data plane's job is to make single-gateway compromise insufficient; correlation-resistance is DAITA's job [01-DP §9].
2. **Jurisdiction diversity is an input the data plane cannot attest** (M6, §11.4.6). The require_distinct_jurisdiction knob makes Entry==Exit *loud* (teardown) but cannot prove the labels are honest. The honest boundary MUST be documented to the operator: "the data plane chains what the control plane assigns; jurisdiction trust is a control-plane + operator responsibility."
3. **Per-hop keys never cross hops** (M2). Enrollment generates a *distinct* WG keypair per hop relationship on the device; private keys never leave the device [research-pki_pq_nat §1.3, 01-DP §4]. The control plane only ever holds the per-hop *public* keys.

4. **No-logging holds per hop** (I5). Each gateway keeps only aggregate counters; a hop chain adds no per-connection table. The `chain_id` in telemetry is aggregate ("chain template X established N times in region R"), never per-user [01-DP §5.3 telemetry rule].
5. **Kill-switch interaction** (referenced — owned by the security doc): an M5 whole-chain teardown **MUST** trip the kill-switch like any tunnel-down event; a half-up chain (forbidden by M5) would be a leak. The state machine's atomic `Failed → TearingDown` is the data-plane guarantee the security doc's kill-switch relies on.
6. **Fail-static** (M5/I3): an established chain keeps forwarding if the control plane goes away; loss of the control plane does not tear down a working chain. Only a *data-leg* failure tears down.
7. **Amplification / resource bound:** `max_hops` (§7) caps chain length so a malicious map cannot request a 1000-hop chain to exhaust client CPU/MTU. `from_spec` rejects over-length chains before any dial.
8. **Post-quantum is additive per hop** (referenced — security doc): the ML-KEM/FIPS-203 PSK [research-pki_pq_nat §2.4] slots into each hop's `Tunn` preshared-key field independently; nesting is unaffected because PQ changes the handshake, not the datagram framing [01-DP §4]. A chain **MAY** mix PQ and classical hops (e.g. PQ on the Entry leg only) — **UNVERIFIED** whether the MVP mandates PQ on all hops; default to all-or-the-security-doc's-call.

12. Test points (§11.4.169 mandatory test types)

Every `hop_chain.rs` workable item declares its required test types from the §11.4.169 closed vocabulary [06-WBS §0.4]. Mapping for this module:

Code	Test point (captured-evidence per §11.4.5/.69/.107)
UT	<code>from_spec</code> validates M2 (<code>duplicate-key → DuplicateHopKey</code>), <code>max_hops</code> , M6 strict (<code>Entry==Exit → JurisdictionNotDistinct</code>); <code>inner_wg_mtu()</code> arithmetic for 1/2/3-hop; encrypt/peel layering order (innermost-first egress, outermost-first ingress); state-machine transition table (every edge in §5.2).
IT	<code>netns rig</code> [06-WBS §0.4]: real Client + real Entry edge + real Exit edge (booted on-demand via containers submodule §11.4.76); 2-hop chain established; assert <code>curl</code> from client reaches the Exit-side target; assert Entry's <code>WgPeer.AllowedIPs</code> for the client = only the Exit overlay (M4 captured from edge config).

Code	Test point (captured-evidence per §11.4.5/.69/.107)
E2E	client overlay namespace → through 2-hop chain → curl http://10.10.0.20/ succeeds; status trace shows Connecting→Handshaking→Connected{transport:"...→..."}.
SEC	the entry/exit-separation proof: tcpdump on the Entry↔Exit leg shows only opaque WG datagrams (Ed) — assert the destination IP/plaintext is NOT recoverable at Entry (M3); tcpdump at Exit shows source = Entry, never the client IP (M4); a key-isolation test asserts Entry holds no κ_{exit} (M2). DPI/no-plaintext-leak per [06-WBS §0.4 SEC].
SC	stress+chaos (§11.4.85): kill the Exit transport mid-transfer → assert WHOLE chain tears down (M5, no half-tunnel) and Down emitted within hop_keepalive_grace; flap the Entry leg → assert atomic teardown, no leaked forwarding.
CONC/ RACE	three-loop orchestrator driving a chain under concurrent egress/ingress; loom/-race on the atomic Established→Failed→TearingDown transition — no torn state where some hops forward and others are dead.
FA	full-automation deterministic (§11.4.50): N=3 identical one-shot runs of establish→transfer→teardown produce identical status traces + identical evidence hashes.
BENCH	bench.sh 2-hop vs 1-hop: RTT (assert $\approx r_1+r_2$), throughput, client CPU/Gbps (assert $\approx 2 \times$ WG ops), MTU measured not assumed (§6 row 3), p99 latency CSV [01-DP §10/§12].
MEM	RSS sampling of the chain forwarder under sustained transfer (no per-packet alloc — §10 budget); the iOS NE memory ceiling (G3 [01-DP §13]) bounds the client chain state.
CH/ HQA	per-gate Challenge scoring the captured SEC/E2E evidence (entry/exit separation proven, not asserted); autonomous HelixQA session against the 2-hop slice [06-WBS §0.4].

The ONLY permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason (e.g. MEM on a host with no iOS device → topology_unsupported), never a silent gap [06-WBS §0.4]. Four-layer enforcement per §11.4.4(b) applies to every TASK closure.

13. Phase placement & checklist

Item	Owns	Phase	Gate
helix-core/src/ hop_chain.rs —		2	— (Phase-2

Item	Owns	Phase	Gate
HopChain/Hop/state machine/ encapsulation	nested-WG egress/ ingress, establish, MTU		BENCH/ SEC)
helix-edge ChainForwarder (§4)	Entry/ Intermediate hop forwarding (M3)	2	—
helix-route HopChainSpec map field (§8)	chain delivery via WatchNetworkMap	2	—

Multi-hop is **Phase 2 (Parity + Reach)** [SYNTHESIS §4, 01-DP §12]. It is purely additive over the frozen Phase-0 interfaces (the Transport trait §[01-DP §3.1], helix-wg §[01-DP §4], the TunnelStatus enum §[01-DP §5.2], the RouteMap §[01-DP §6.2]) — it adds one struct module and one map field, **no new wire protocol** (M1).

13.1 Anti-bluff acceptance evidence (§11.4.5/.69/.107)

The release-blocking captured evidence for multi-hop:

- **Entry/exit-separation capture** (the SEC test above): packet captures at Entry and at Exit, plus the key-isolation assertion, proving M2/M3/M4 — not a config-only PASS.
- **Whole-chain teardown trace**: ordered TunnelStatus events proving M5 atomicity under the SC fault injection.
- **2-hop BENCH CSV**: measured RTT/throughput/CPU/MTU vs single-hop, so the latency cost is real and quantified, not assumed (§11.4.6).

End of Multi-Hop (Entry/Exit Separation) nano-detail specification. Pairs with doc 01-DP §11.1 (the section it deepens), doc 03 (WatchNetworkMap — the live source of the HopChainSpec stream at §8), the security volume (kill-switch/DNS-leak §11.5, post-quantum PSK §11.8), and doc 02 (control-plane chain & jurisdiction assignment policy — M6).